What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Nonbacterial Thrombotic Endocarditis

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Adapted from: Nonbacterial Thrombotic Endocarditis

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of nonbacterial thrombotic endocarditis (NBTE).
2. Explain the pathophysiology of NBTE.
3. Interpret clin NBTE.
4. Outline the management plan of NBTE.

What condition underlies most cases of NBTE?

How Do Patients with Nonbacterial Thrombotic Endocarditis Present? (LO #1)

Patients with NBTE are often asymptomatic. In fact, it is most often diagnosed at autopsy, detected as an incidental finding. There may be no indications of valvular or heart dysfunction. This is because since the valvular lesions are non-infectious, they rarely result in destruction of the valve, as commonly occurs in infective endocarditis.

When present, symptoms are typically because of **thromboembolism** rather than due to direct valvular damage. In some cases, patients may not be diagnosed until after emboli have occurred. Common sites of embolization include the spleen, kid present as flank pain, hematuria, rash, and the central nervous

Systemic Lupus Erythematosus

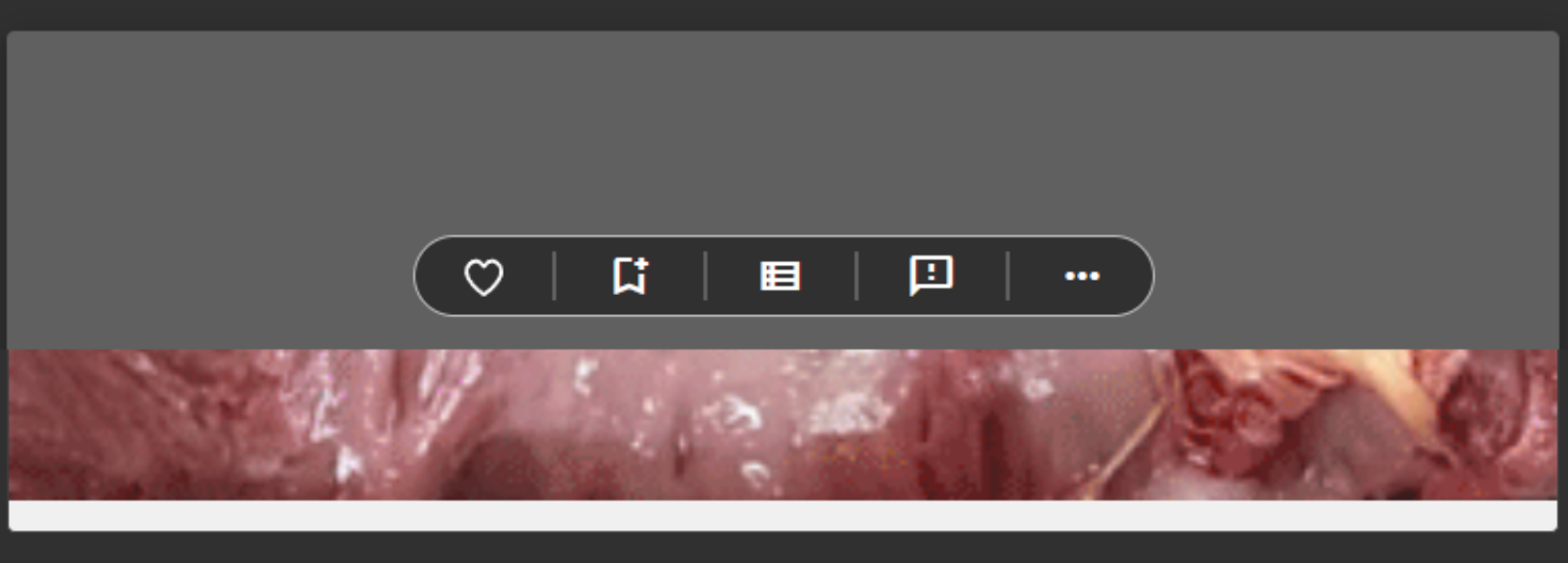
Systemic lupus erythematosus (SLE) is a multisystem autoimmune disease that can show symptoms in many organ systems, including the heart (pericarditis, or less often NBTE). The mechanism of vegetation formation is not well defined. Immune complex formation has been implicated, as the vegetations have been shown to contain immune complexes and cellular debris.

CLINICAL CORRELATION

Some pathologists separate the sterile endocarditis of SLE into a separate diagnosis termed **Libman-Sacks endocarditis**, but this is not uniformly agreed upon. Others use the eponym (Libman-Sacks endocarditis) for a form of NBTE with very large thrombi.

Antiphospholipid Syndrome

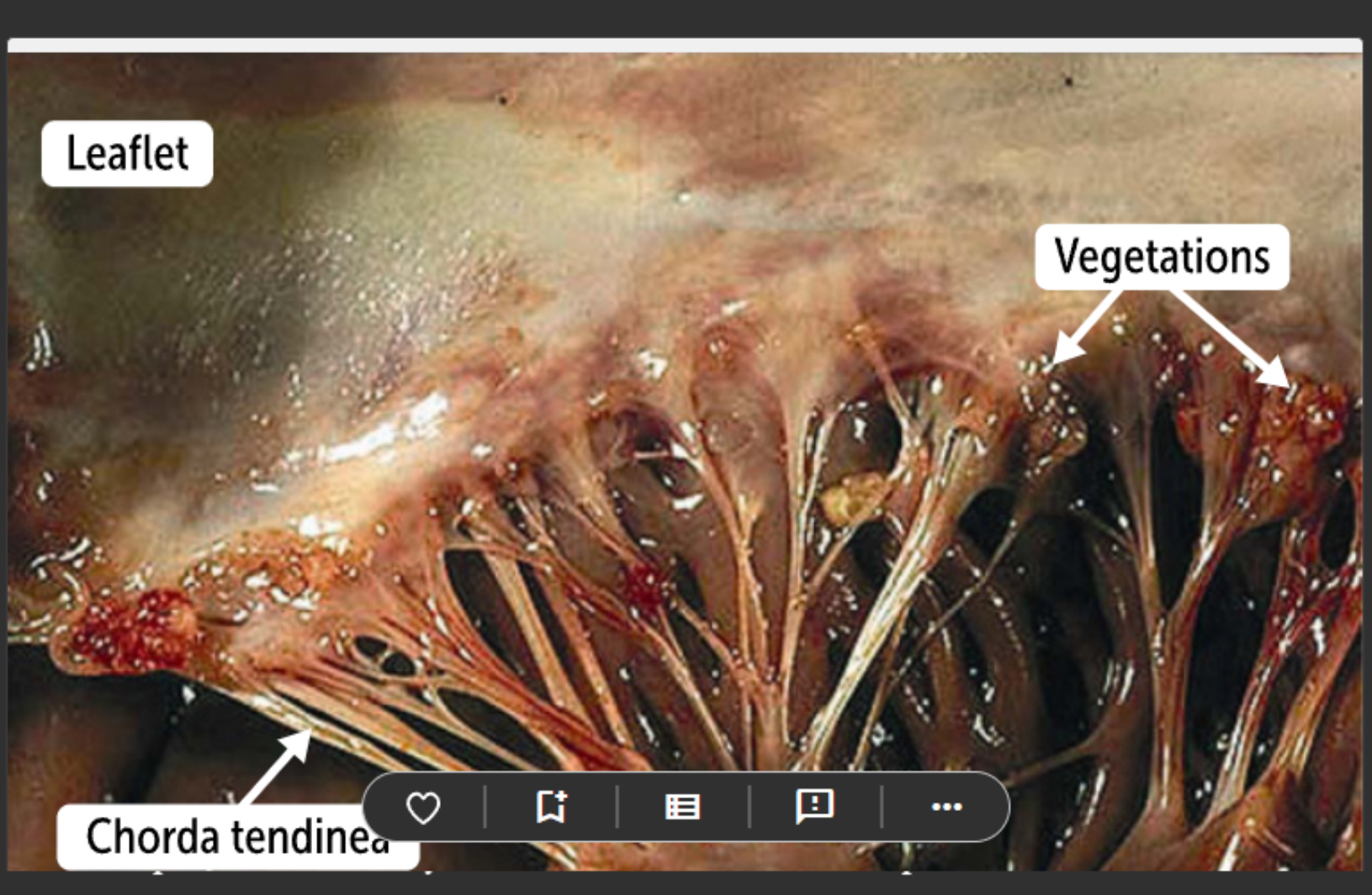
Antiphospholipid syndrome (APS) occurs when autoantibodies attack cell membranes, causing excess clot formation. APS can occur as an idiopathic disorder but also in association with SLE or infections.



QUIZ

Figure 1. Valve Vegetations in NBTE.

The vegetations can occur on any valve, but vegetations associated with SLE favor the mitral and aortic valves. Single or multiple growths can be seen near the margins of the valve cusps. Some large thrombi can form on both sides of the valve, as in SLE endocarditis (Figure 2).



the brain (manifesting as stroke), but the heart, kidney, and spleen are also commonly affected.

The sterile vegetations of NBTE are rarely destructive to the valve, unlike those of infective endocarditis (IE). However, sometimes inflammatory valvulitis can impair valve function, potentially causing valve regurgitation.

The lesions of Libman-Sacks endocarditis characteristically heal with scarring; the ensuing fibrosis can result in lesions that resemble chronic rheumatic heart disease.

What is the most hazardous complication of NBTE?

How Do We Diagnose Nonbacterial Thrombotic Endocarditis? (LO #3)

The diagnosis of NBTE is difficult. Patients are often asymptomatic, and

Why should a patient with a new-onset heart murmur receive an echocardiogram?

Laboratory Tests

Three sets of blood cultures should be done to exclude infective endocarditis. If autoimmune disease (SLE and APS) is suspected as the cause, appropriate laboratory workup such as an antinuclear antibody (ANA), anti-Smith, rheumatoid factor, anti-dsDNA, and antiphospholipid antibody panel should also be done.

How Do We Treat Nonbacterial Thrombotic Endocarditis? (LO #4)

Treatment involves identifying the underlying cause (malignancy, SLE, APS) and treating that if p ts, these patients should be anticoagulated with heparin (unfractionated or low molecular

- contrast with the destructive vegetations of infective endocarditis.
- Vegetations in NBTE may embolize to different parts of the body.
- Diagnosis is made by echocardiography.
- Blood cultures negative for infective endocarditis support the diagnosis.
- Treatment is of the underlying cause (eg, SLE, APS, malignancy).
- Anticoagulation should be used if no contraindication exists.

Review Questions

1. Which of the following mechanisms most commonly leads to nonbacterial thrombotic endocarditis?

- **A.** ○ Autoimmune response against the pericardium
- **B.** ○ Colonization of the heart valve by *Staphylococcus aureus*
- **C.** ● Hypercoagulable state
- **D.** ○ Malignant cancer metastasizing to the heart valves
- **E.** ○ Structural valvular disease

Hide Explanation

The correct answer is hypercoagulable state (C). An underlying hypercoagulable state, leading to wide most patients with nonbacterial thrombotic endocarditis (NBTE). An autoimmune response against the (E) may cause valvular damage and can set up the scene for the formation of an infectious vegetation.

3. A patient previously diagnosed with systemic lupus erythematosus comes to the clinic with history of sudden-onset weakness of her right leg. A murmur is auscultated on cardiac examination that was not documented in previous visits. What is the most likely complication that this patient may have developed?

- **A.** ○ Acute infective endocarditis
- **B.** ● Brain embolization
- **C.** ○ Cardiomyopathy
- **D.** ○ Septic shock
- **E.** ○ Valvular degeneration

Hide Explanation

The correct answer is brain embolization (B). This patient has systemic lupus erythematosus, a predisposing condition to nonbacterial thrombotic endocarditis (NBTE). This, associated with the presence of a heart murmur, signifies that she has NBTE. Since the vegetations of NBTE are only loosely adherent to the valves, they are more prone to det at risk for acute infective endocarditis cess. with diagnosing valvular vegetations.

5. The management of NBTE involves treatment meant to prevent which known complication?

- **A.** ○ Diastolic heart failure
- **B.** ○ Valve destruction
- **C.** ● Embolization
- **D.** ○ Myocarditis
- **E.** ○ Pleural effusion

Hide Explanation

The correct answer is embolization (C). Systemic embolization is the most common and clinically significant complication of NBTE. One of the most common places for embolization is the brain (manifesting as stroke), but the heart, kidney, and spleen are also commonly affected. Unless a contraindication exists, patients with NBTE should be treated with anticoagulants to prevent embolization. Importantly, the vegetations of NBTE are nondestructive, which is in contrast to the valve destruction (B) characteristic of the vegetations associated with infective endocarditis. Diastolic heart failure (A), myocarditis (D), and pleural effusions are not common or significant complications associated with NBTE.

- https://www.sciencedirect.com/science/article/abs/pii/S0033062022001086
- https://www.uptodate.com/contents/nonbacterial-thrombotic-endocarditis?search=nonbacterial%20thrombotic%20endocarditis&source=search_result&selectedTitle=1~28&usage_type=default&display_rank=1
- Kumar V, Abbas AK, Aster JC, eds. *Robbins Basic Pathology*. 10th ed. Elsevier; 2018.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☑ ~~Differentiate the clinical presentation of nonbacterial thrombotic endocarditis (NBTE).~~
- 02 ☑ ~~Explain the pathophysiology of NBTE.~~
- 03 ☑ ~~Interpret clinical findings to properly diagnose NBTE.~~
- 04 ☑ ~~Outline the management plan of NBTE.~~

Objective Confidence: 4/4